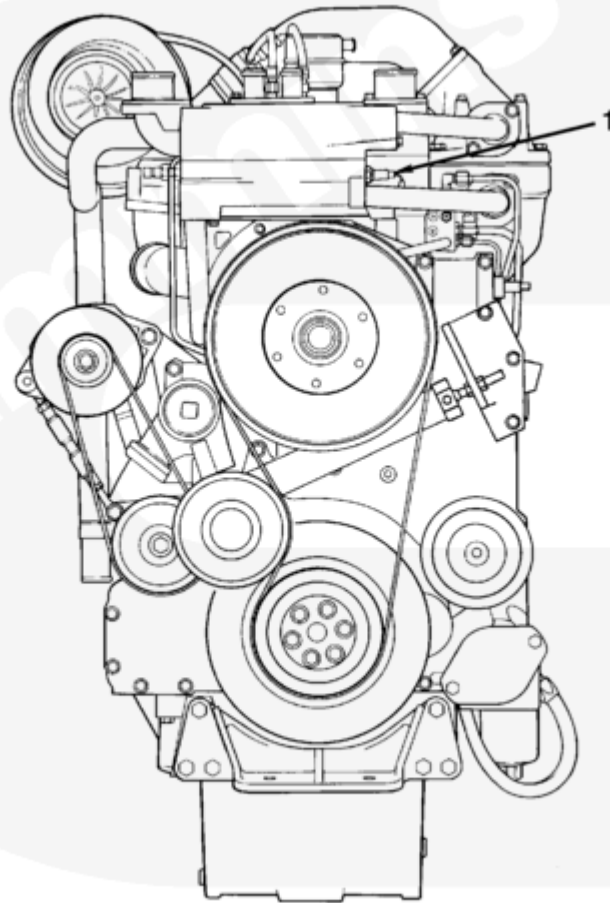


Trainings ▾

Engine Views

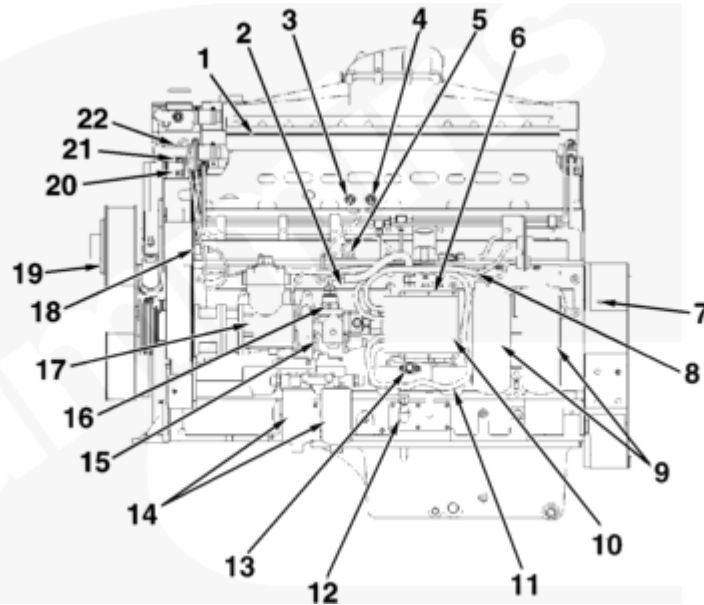
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19400097

QSK19 - Front View

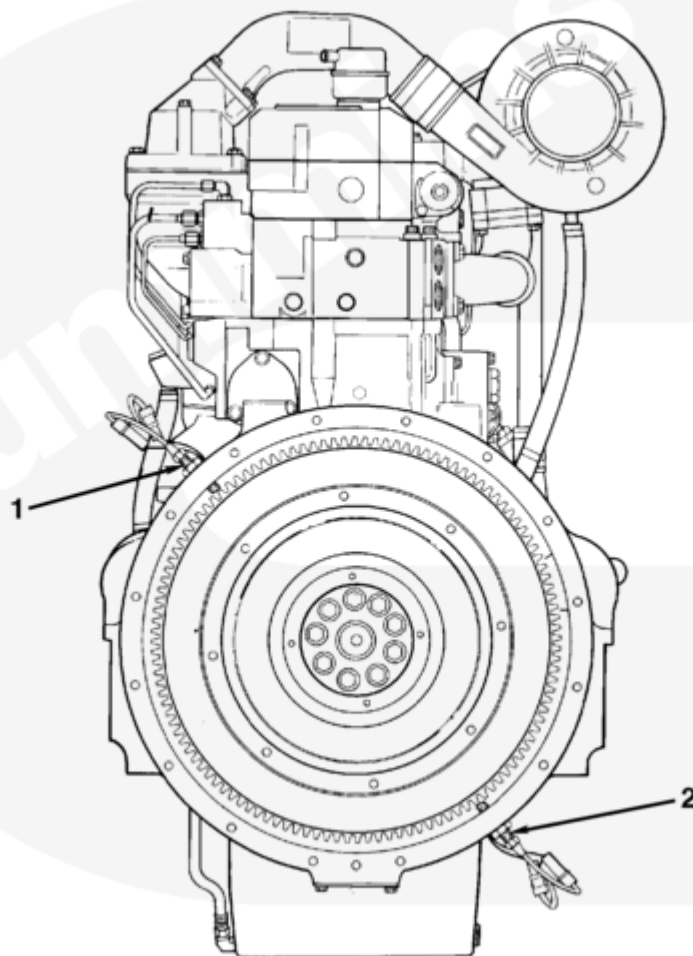
1. Coolant temperature sensor.



00400144

QSK19 - Intake Side View

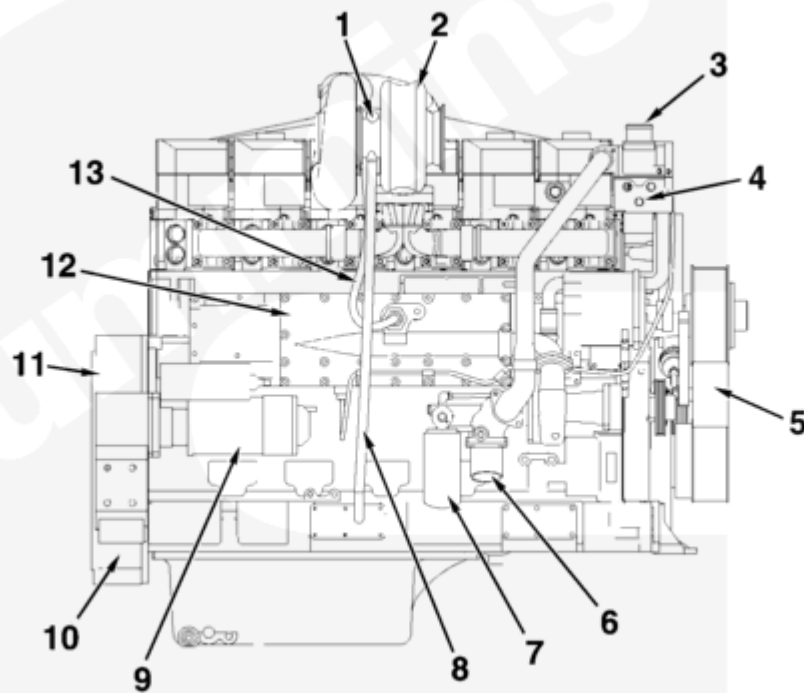
1. Aftercooler assembly
2. Oil pressure sensor
3. Intake manifold temperature sensor
4. Intake manifold pressure sensor
5. Datalink connector
6. Control valve body
7. Engine speed sensor (generator drive)
8. OEM interface harness
9. Combination full flow/bypass oil filters
10. Electronic control module (ECM)
11. Engine harness
12. Dipstick
13. Ambient air pressure sensor
14. Fuel filters
15. Fuel pump
16. Fuel pump outlet quick-disconnect fitting
17. Air compressor
18. Engine speed sensor (**not** generator drive)
19. Fan hub
20. Fuel rail quick-disconnect fitting
21. Timing rail quick-disconnect fitting
22. Coolant temperature sensor.



19400098

QSK19 - Rear View

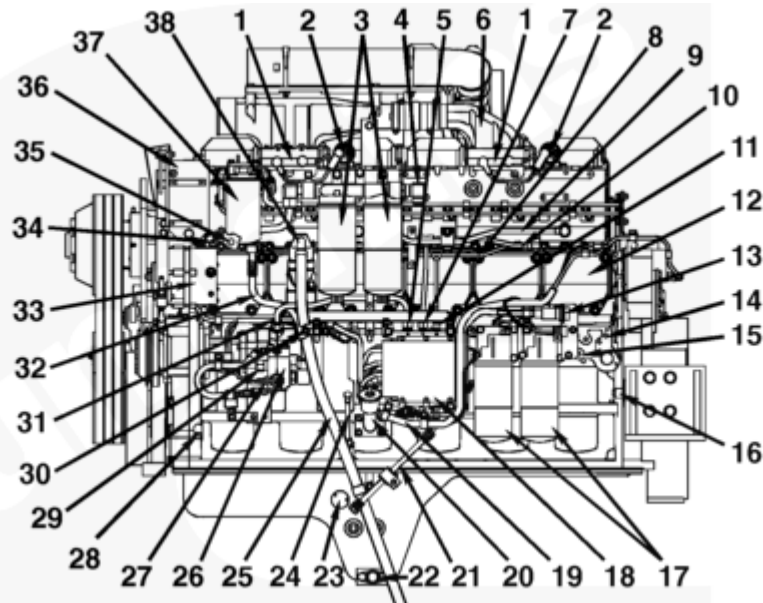
1. Engine speed sensor (generator drive)
2. Alternative engine speed sensor (generator drive).



19400997

QSK19 - Exhaust Side View

1. Oil inlet to turbocharger
2. Turbocharger
3. Coolant outlet
4. Thermostat housing
5. Fan idler and assembly
6. Coolant inlet
7. Coolant filter
8. Turbocharger oil drain
9. Starting motor
10. Alternate location for engine speed sensor
11. Flywheel housing
12. Oil cooler
13. Coolant supply to turbocharger.



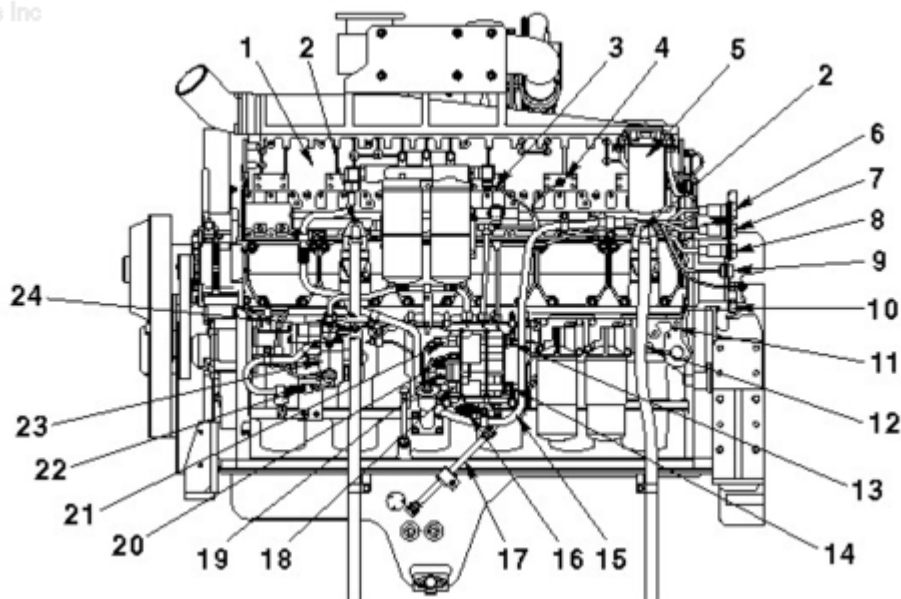
00400154

QSK23 Industrial - Left Side

1. Grid heaters (Industrial **only**)
2. Grid heater relays (Industrial **only**)
3. Fuel filters
4. Fuel inlet (to filters)
5. Fuel inlet (to ECVA)
6. Air intake connection (Industrial **only**)
7. Fuel timing supply
8. Fuel pipe (to fuel rail)
9. Fuel pipe (return)
10. Fuel pipe (for injection timing)
11. Fuel rail supply
12. Cam follower cover
13. OEM electrical connections (Industrial **only**)
14. Oil pressure check point (after filters)
15. Oil pressure check point (before filters)
16. Engine barring device
17. Oil filters
18. ECVA/ECM
19. Wiring harness
20. Oil fill port
21. Fuel rail dampening hose
22. Engine oil drain valve
23. Oil level sensor (optional)
24. Dipstick
25. Blowby tube
26. Fuel pump
27. Fuel pump pressure sensor
28. Oil pressure check point (to front idler gears)
29. Fuel pump actuator

30. Oil pressure sensor
31. Fuel pipe (pump to ECVA)
32. Fuel pipe (filters to pump)
33. Alternator
34. Intake manifold pressure sensor (Industrial **only**)
35. Intake manifold temperature sensor (Industrial **only**)
36. Freon compressor mounting location
37. Coolant filter (Industrial **only**)
38. Crankcase breather assembly.

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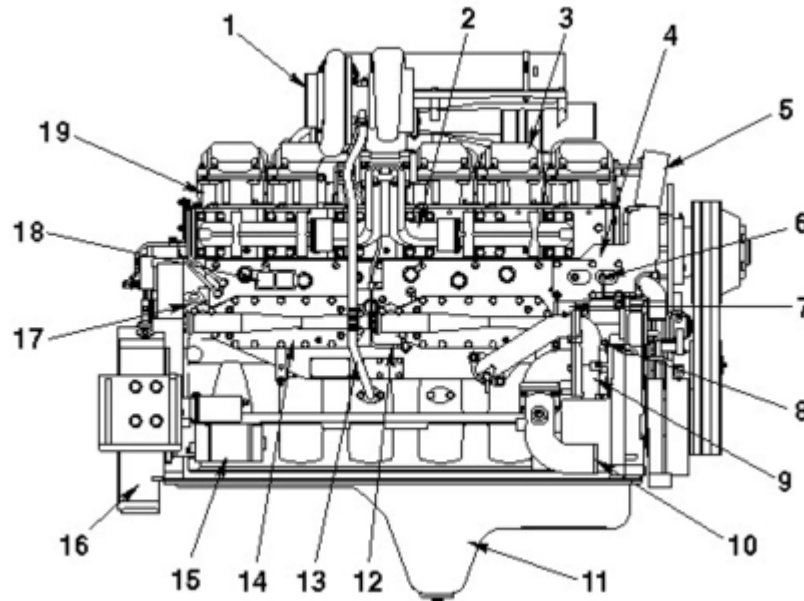
00400162

QSK23 Power Generation - Left Side

1. Intake manifold (Power Generation **only**)
2. Crankcase breathers
3. Intake manifold pressure sensor (Power Generation **only**)
4. Intake manifold temperature sensor (Power Generation **only**)
5. Coolant filter (Power Generation **only**)
6. Inline B connector (Power Generation **only**)
7. Inline A connector (Power Generation **only**)
8. Inline C connector (Power Generation **only**)
9. Datalink connector (Power Generation **only**)
10. Engine speed sensor (Power Generation **only**)
11. Oil pressure check point (after filters)
12. Oil pressure check point (before filters)
13. Fuel timing pressure sensor
14. Fuel rail pressure sensor
15. Wiring harness
16. Barometric pressure sensor
17. Fuel rail dampening hose
18. Fuel rail actuator
19. Fuel shutoff valve

- 20. Fuel temperature sensor
- 21. Fuel timing actuator
- 22. Fuel pump pressure sensor
- 23. Fuel pump actuator
- 24. Oil pressure sensor.

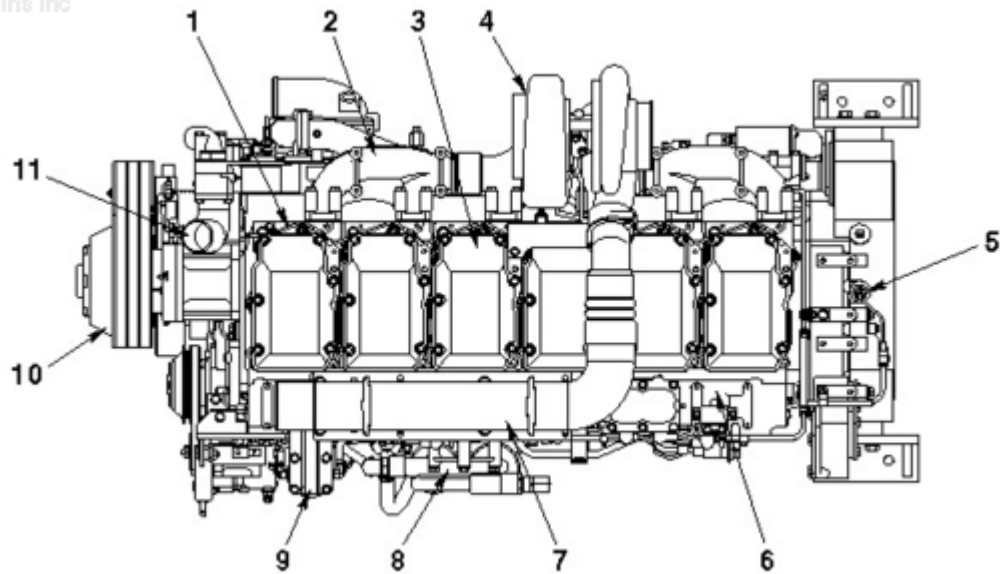
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00400165

QSK23 - Right Side

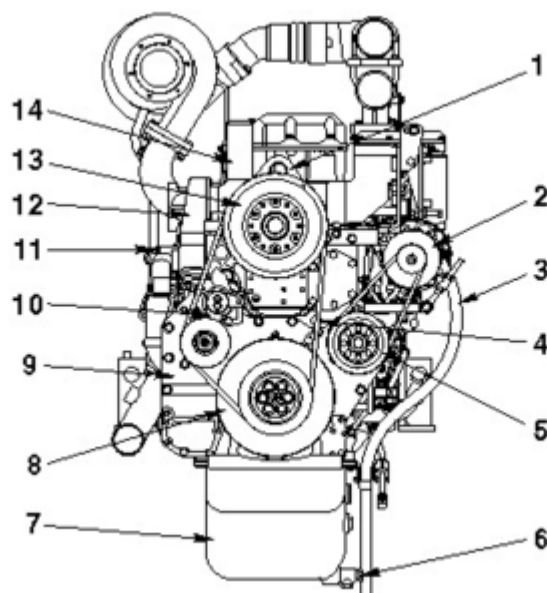
- 1. Turbocharger
- 2. Exhaust manifold (3-piece)
- 3. Rocker lever cover
- 4. Thermostat housing
- 5. Engine coolant outlet
- 6. Coolant temperature sensor
- 7. Coolant bypass tube
- 8. Oil pressure checkpoint (piston cooling nozzle gallery)
- 9. Water pump
- 10. Engine coolant inlet
- 11. Oil pan
- 12. Turbocharger oil supply line
- 13. Turbocharger oil drain line
- 14. Oil cooler housing
- 15. Starter
- 16. Flywheel housing
- 17. Oil pressure checkpoint (after oil coolers)
- 18. Engine dataplate
- 19. Rocker lever housing.



00400166

QSK23 - Top View

1. Coolant vent tube
2. Exhaust manifold
3. Rocker lever cover
4. Turbocharger
5. Engine speed sensor (Industrial **only**)
6. Intake manifold
7. Charge air piping (turbocharger to CAC)
8. Fuel filter head
9. Coolant filter head (Industrial **only**)
10. Fan hub assembly
11. Engine coolant outlet.

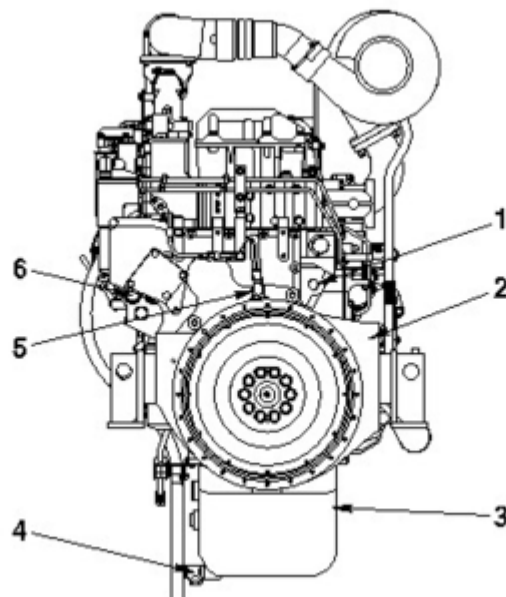


00400167

QSK23 - Front View

1. Front engine lifting bracket
2. Alternator
3. Blowby tube
4. Cooling fan drive belt
5. Drive pulley (for alternator and Freon compressor)
6. Oil drain valve
7. Oil pan
8. Vibration damper
9. Front gear cover
10. Cooling fan belt tensioner (Industrial **only**)
11. Coolant temperature sensor
12. Thermostat housing
13. Fan hub assembly
14. Engine coolant outlet.

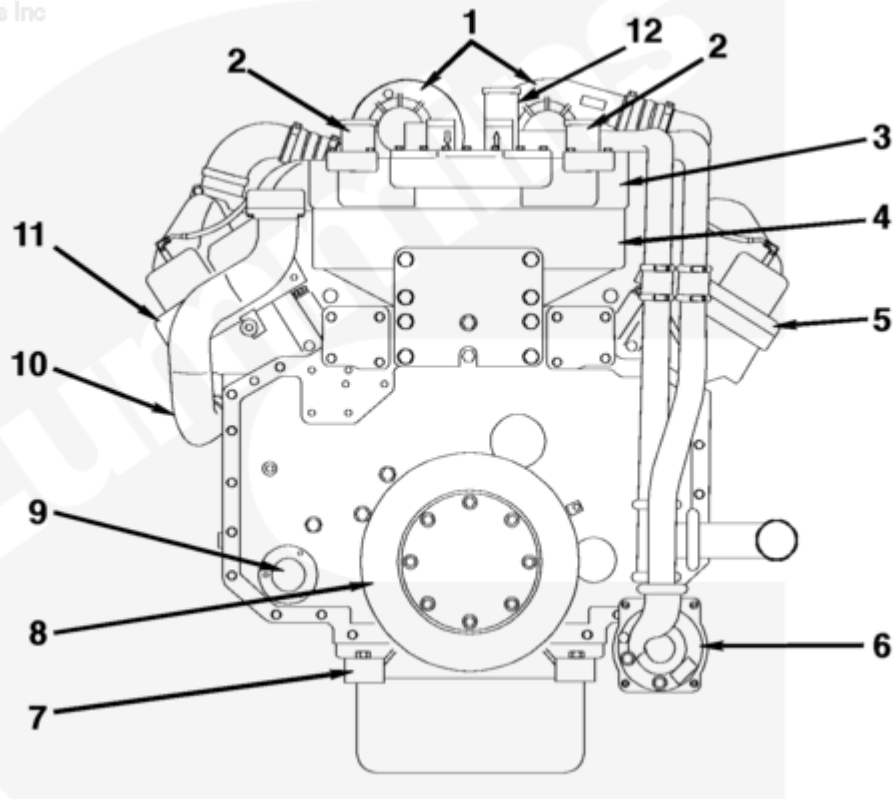
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00400168

QSK23 - Rear View

1. Oil pressure check point (after oil coolers)
2. Flywheel housing
3. Oil pan
4. Engine oil drain valve
5. Engine speed sensor (Industrial **only**)
6. OEM electrical connections (Industrial **only**).

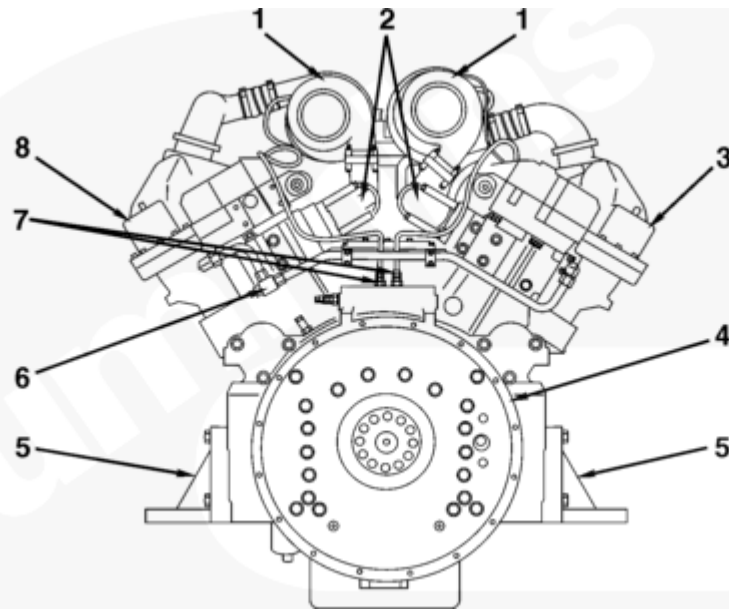


19400998

QSK45 - Front View

1. Turbochargers
2. Engine water outlet connections (non-LTA)
3. Thermostat housing
4. Thermostat housing support
5. Left-bank aftercooler assembly
6. LTA water pump
7. Front engine mount
8. Vibration damper
9. Alternative drive
10. Water bypass tube
11. Right-bank aftercooler assembly
12. LTA water outlet connection.

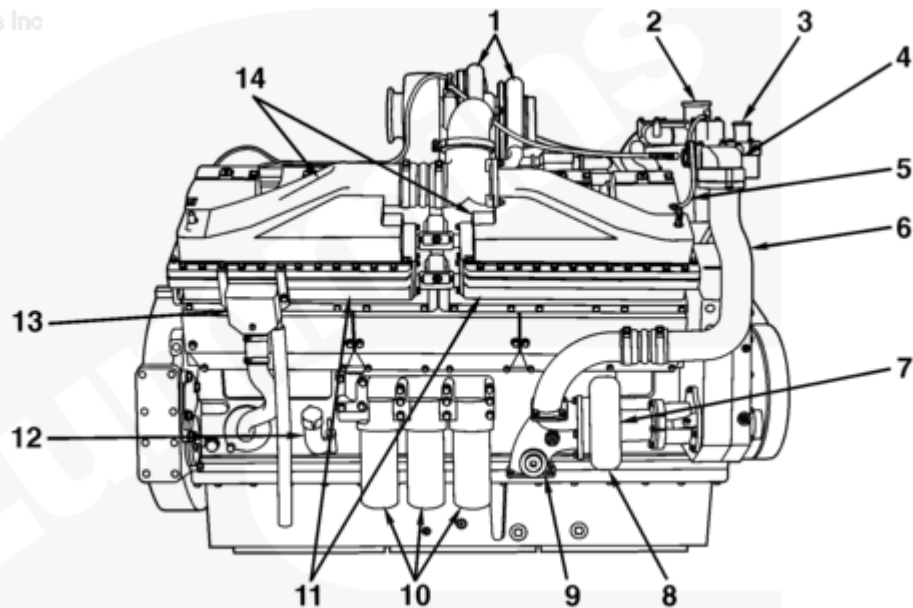
Note : Some engine configurations have the fuel return lines positioned at the front of the engine.



00600017

QSK45 - Rear View

1. Turbochargers
2. Exhaust manifolds
3. Right-bank aftercooler assembly
4. Flywheel housing
5. Rear engine mount
6. Fuel return line
7. Turbocharger oil supply lines
8. Left-bank aftercooler assembly.



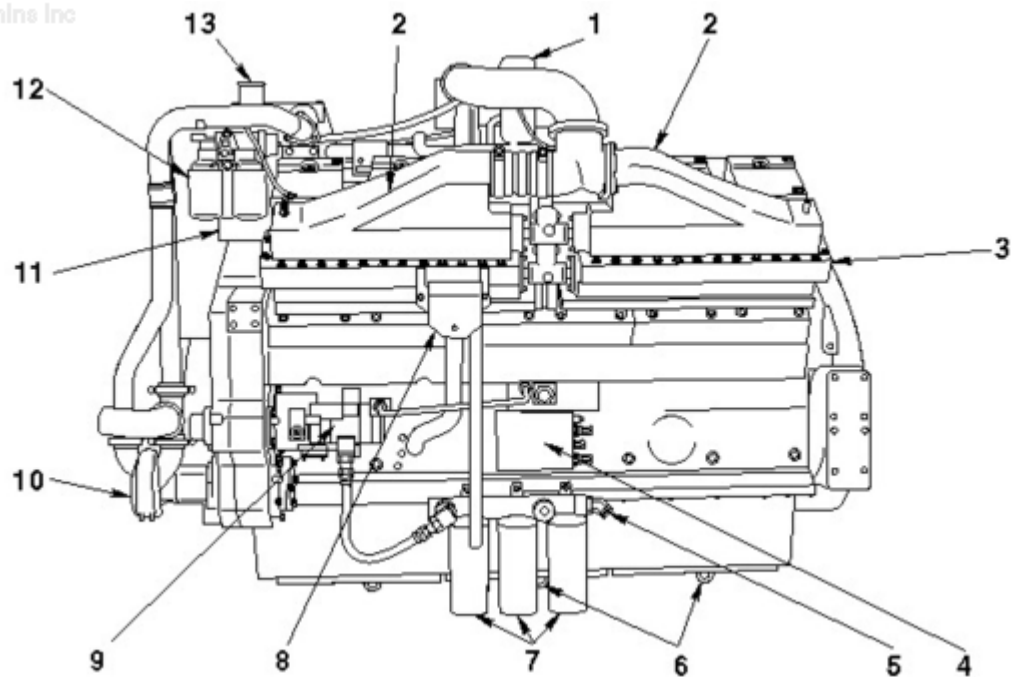
00600013

QSK45 - Right View

1. Turbochargers
2. LTA water outlet connections
3. Engine water outlet connections

4. Engine thermostat outlet
5. Aftercooler vent line
6. Water bypass tube
7. Water pump
8. Water pump drain
9. Coolant inlet connection
10. Oil filters
11. Aftercooler assemblies
12. Oil fill and dipstick tubes
13. Crankcase breather
14. Aftercooler cover.

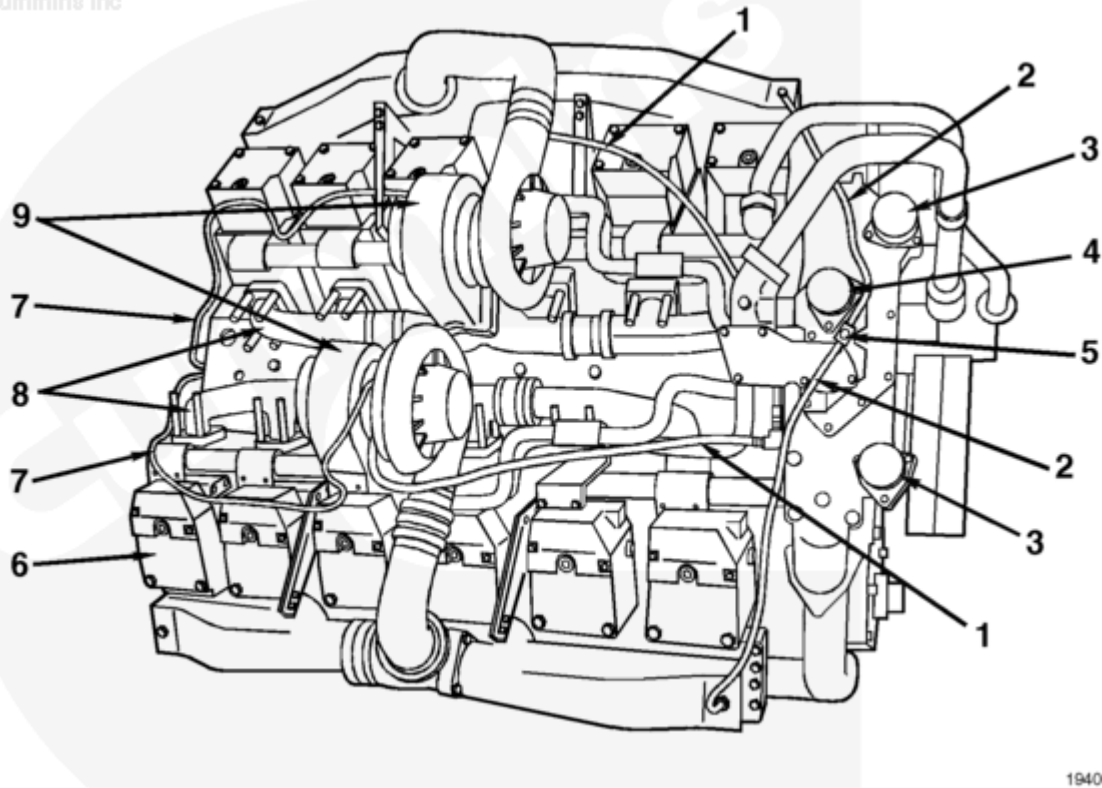
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00600014

QSK45 - Left View

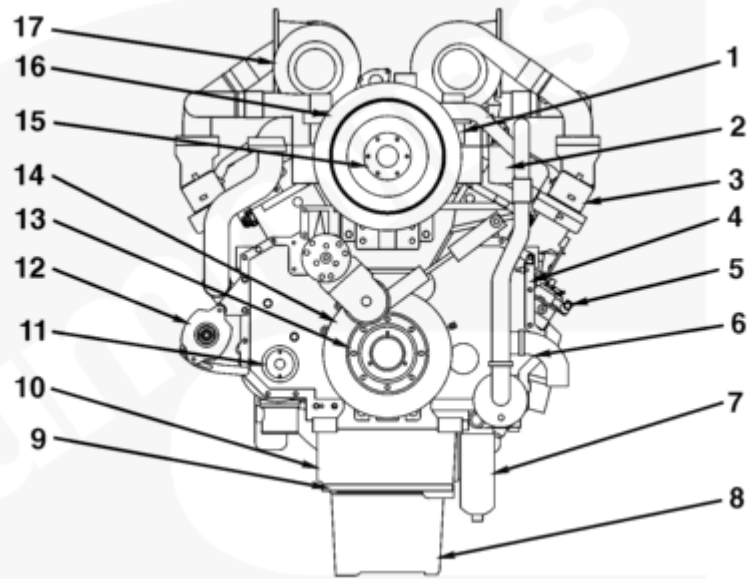
1. Turbocharger
2. Aftercooler covers
3. Aftercooler assemblies
4. Electronic control module (ECM)
5. Fuel inlet connection
6. Engine oil drains
7. Fuel filters
8. Crankcase breather
9. Fuel pump
10. LTA water pump
11. Thermostat housing support
12. Water filters
13. LTA water outlet connections.



19400999

QSK45 - Top View

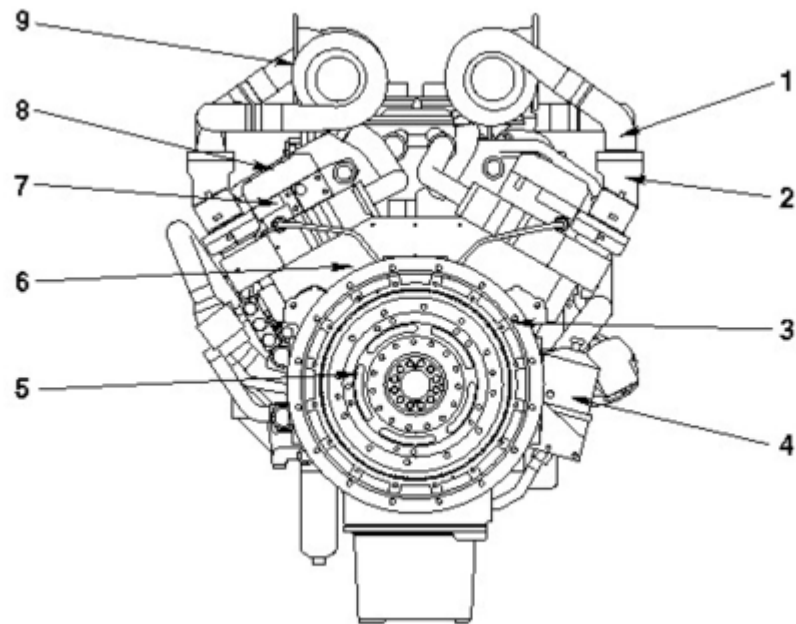
1. Coolant supply lines to turbocharger
2. Coolant vent connection aftercoolers
3. Engine water outlet connection
4. LTA water outlet connection
5. Aftercooler vent line connection engine
6. Rocker lever cover
7. Oil supply lines to turbocharger
8. Exhaust manifolds
9. Turbochargers.



00600021

QSK60 - Front View

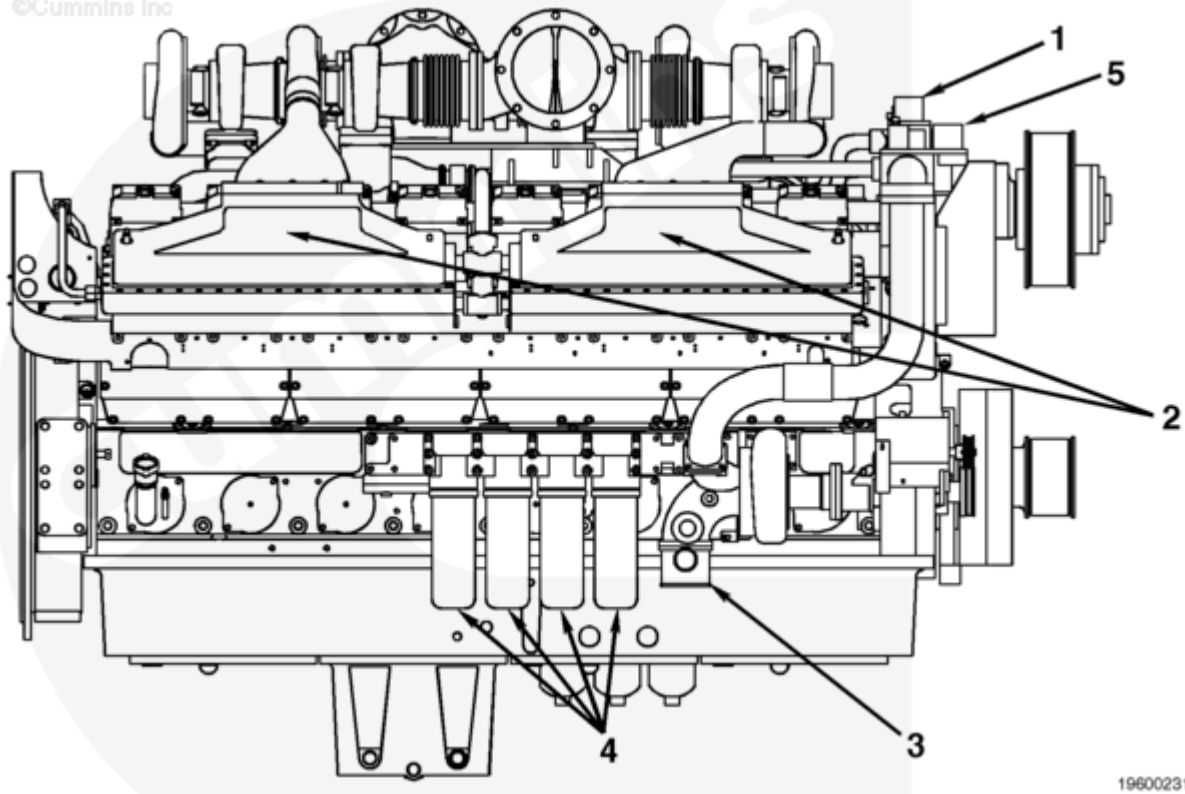
1. Thermostat housing
2. Coolant filter
3. Aftercooler assembly
4. Gear cover
5. Air compressor
6. Accessory drive pulley location (**not** shown)
7. Fuel filter
8. Oil pan sump
9. Oil pan adapter plate
10. Oil pan adapter
11. Alternator drive pulley
12. Alternator
13. Crankshaft pulley
14. Vibration dampers
15. Fan hub
16. Fan hub pulley
17. Turbocharger.



00600022

QSK60 - Rear View

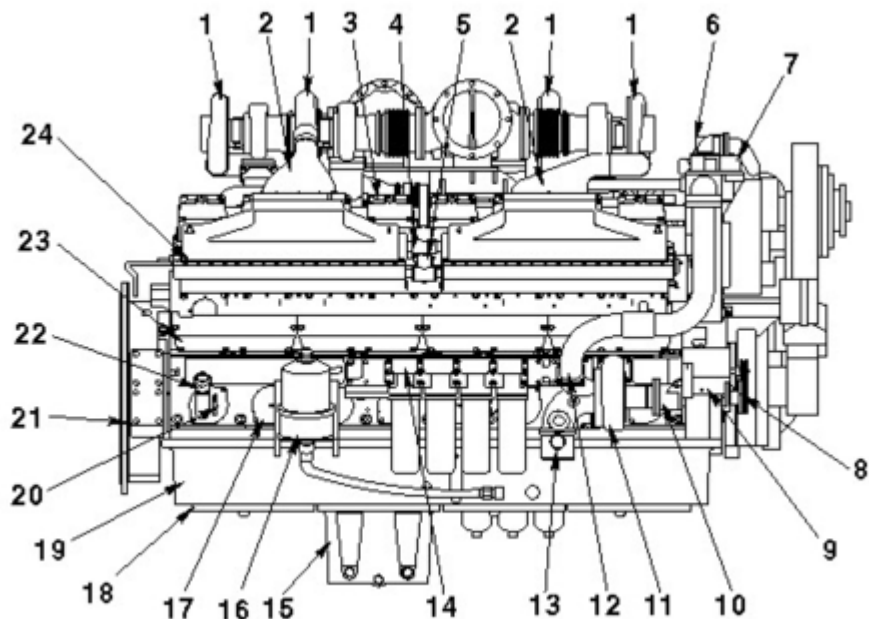
1. Air crossover connector
2. Aftercooler assembly
3. Flywheel housing
4. Centrifuge bypass filter
5. Flex plate
6. Cylinder block
7. Rocker lever housing
8. Rocker lever cover
9. Turbocharger.



19600231

QSK60 - Right View

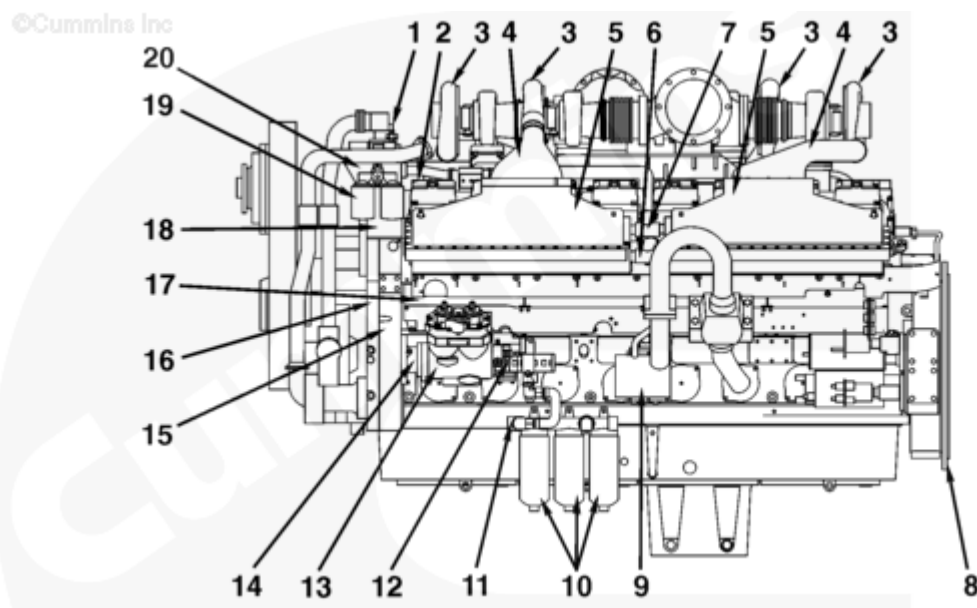
1. LTA coolant outlet
2. Aftercooler assembly
3. Coolant inlet
4. Oil filters
5. Engine coolant outlet.



00600018

QSK60 - Right View

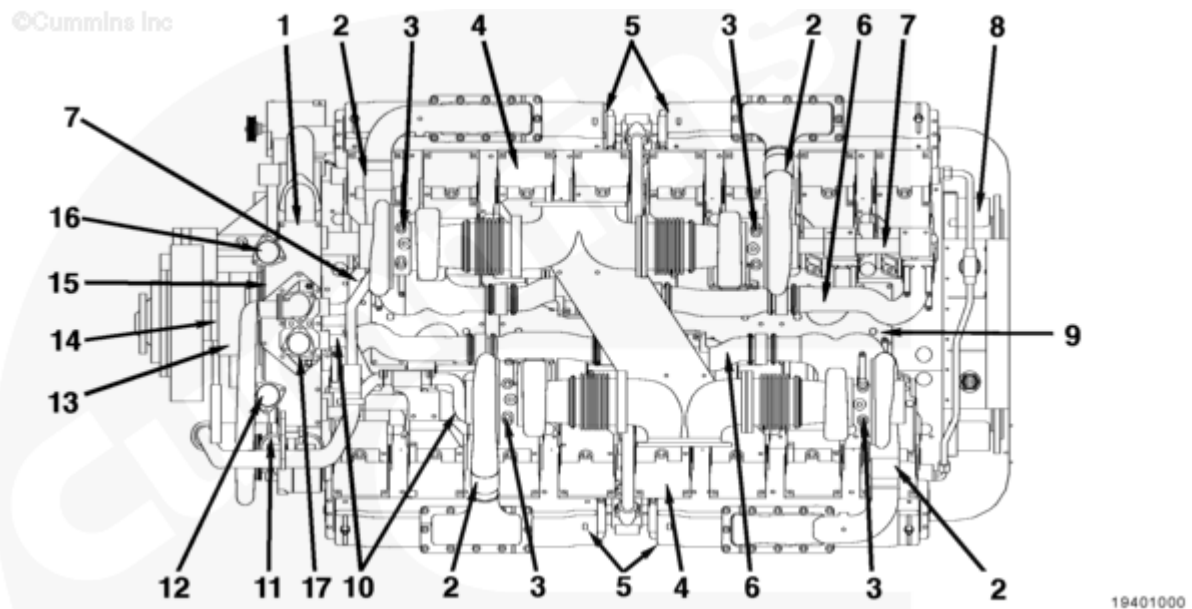
1. Turbochargers
2. Air crossover connections
3. Rocker lever cover
4. Aftercooler water outlet tube
5. Aftercooler water inlet tube
6. LTA Water outlet connections
7. Engine water outlet connections
8. Alternator drive
9. Alternator
10. Water pump drive
11. Water pump
12. Water bypass tube
13. Water inlet connection
14. Oil filter head
15. Oil pan sump
16. Centrifuge bypass filter
17. Hand-hole covers
18. Oil pan adapter plate
19. Oil pan adapter
20. Dipstick
21. Flywheel housing
22. Oil fill tube
23. Cam follower cover
24. Aftercooler assembly.



QSK60 - Left View

1. LTA water outlet connections
2. Rocker lever cover
3. Turbochargers
4. Air crossover connection

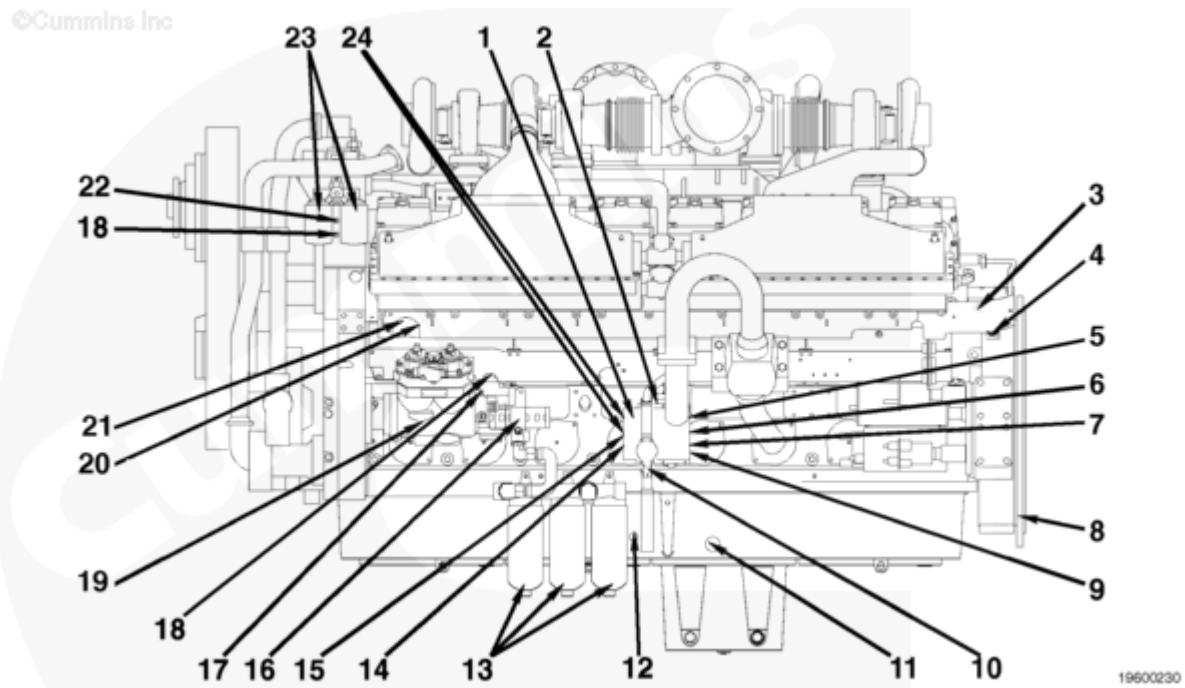
5. Aftercooler assemblies
6. Aftercooler water inlet tube
7. Aftercooler water outlet tube
8. Flywheel housing
9. Electronic control module (ECM)
10. Fuel filter
11. Fuel filter head
12. Fuel pump
13. Air compressor
14. Accessory drive
15. Front gear housing
16. Front gear cover
17. Cam follower cover
18. Thermostat housing support
19. Coolant filters
20. Thermostat housing.



QSK60 - Top View

1. Thermostat housing
2. Air crossover connections
3. Turbochargers
4. Rocker lever cover
5. Aftercooler assemblies
6. Exhaust manifolds
7. LTA water tube
8. Flywheel housing
9. Oil cooler cover plate
10. Aftercooler water tube
11. Coolant filter head
12. Engine coolant outlet (non-LTA)

- 13. Fan hub support
- 14. Fan hub shaft
- 15. Thermostat housing support
- 16. Engine coolant outlet (non-LTA)
- 17. LTA coolant outlet.

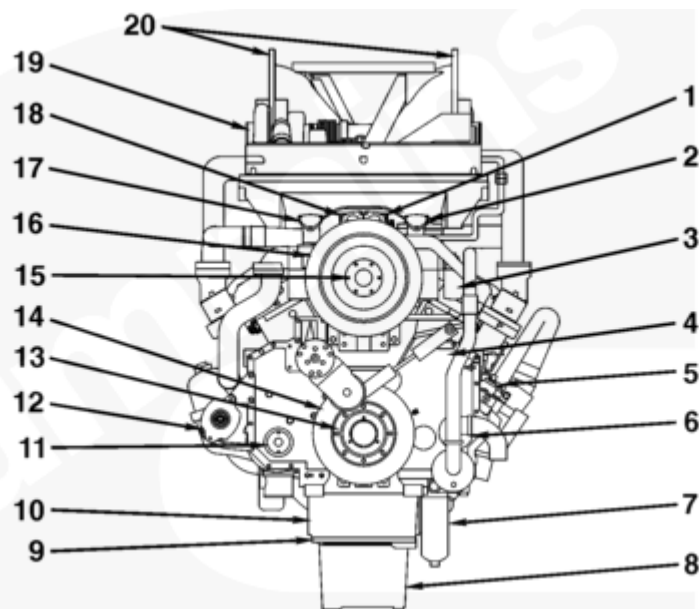


QSK60 (QSK45 similar) - Top View (sensor locations)

- 1. Electronic control module (ECM)
- 2. Control valve body
- 3. Oil pressure sensor
- 4. Engine speed sensor
- 5. Timing pressure sensor
- 6. Barometric pressure sensor
- 7. Fuel temperature sensor
- 8. Flywheel housing
- 9. Rail pressure sensor
- 10. Blowby sensor
- 11. Oil level sensor
- 12. Oil temperature sensor
- 13. Fuel filters
- 14. Rail actuator
- 15. Fuel shutoff valve
- 16. Fuel pump
- 17. Fuel pump quick-disconnect fitting
- 18. Coolant pressure sensor
- 19. Air compressor
- 20. Intake manifold temperature sensor
- 21. Intake manifold pressure sensor

- 22. Coolant temperature sensor
- 23. Coolant filters
- 24. Timing actuators.

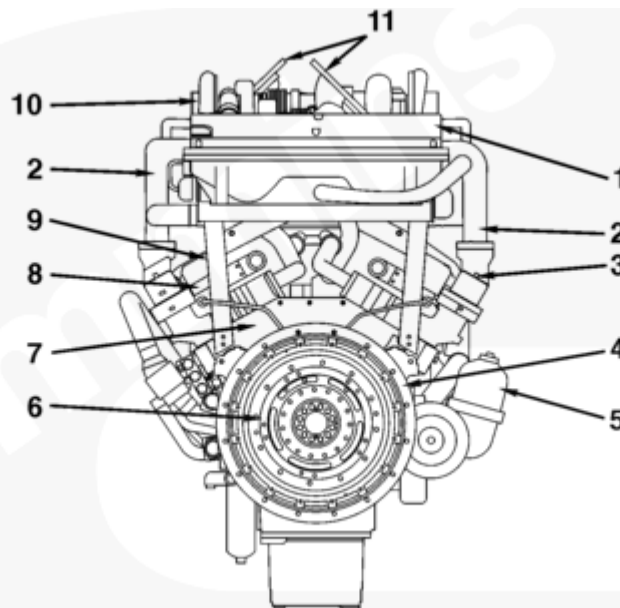
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00600026

QSK60, 2-Stage - Front View

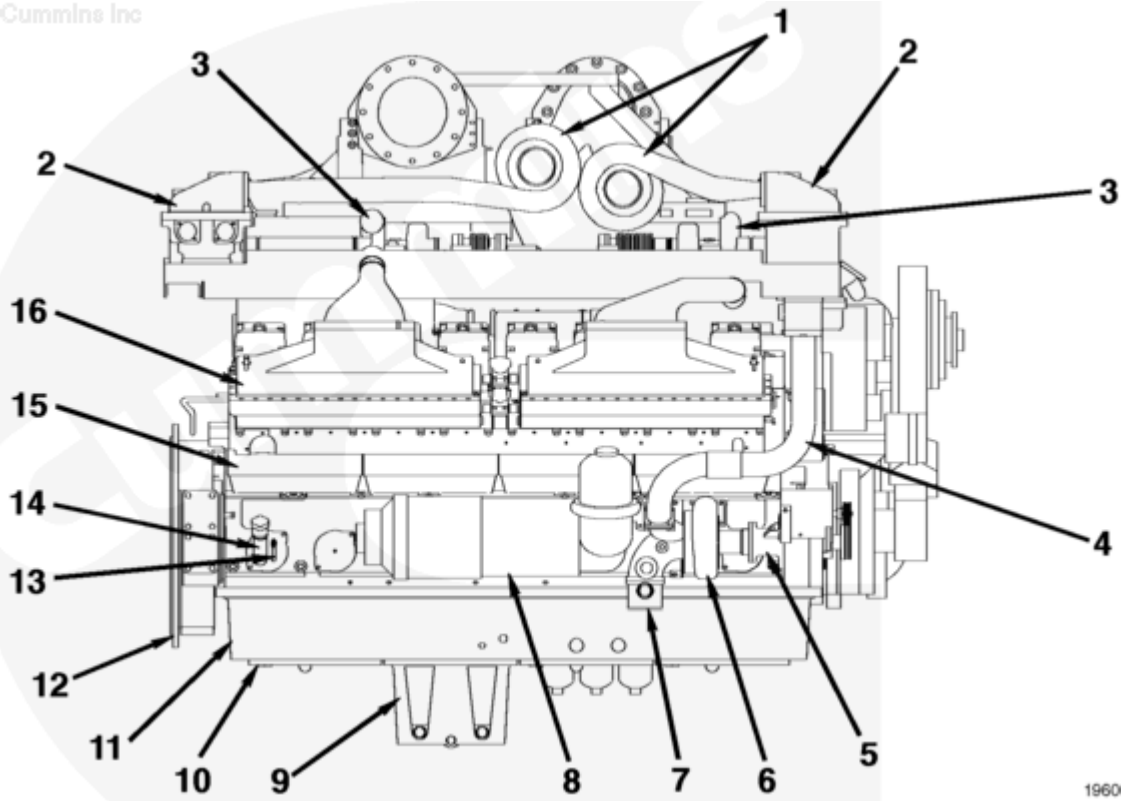
- 1. LTA water inlet connection
- 2. Left-bank water outlet connection
- 3. Coolant filters
- 4. Gear cover
- 5. Air compressor
- 6. Accessory drive pulley
- 7. Fuel filters
- 8. Oil pan sump
- 9. Oil pan adapter plate
- 10. Oil pan adapter
- 11. Alternator drive pulley
- 12. Alternator
- 13. Crankshaft pulley
- 14. Vibration damper
- 15. Fan hub
- 16. Fan hub pulley
- 17. Right-bank coolant outlet connection
- 18. LTA water outlet connection
- 19. Turbocharger
- 20. Exhaust flange outlets.



00600027

QSK60, 2-Stage - Rear View

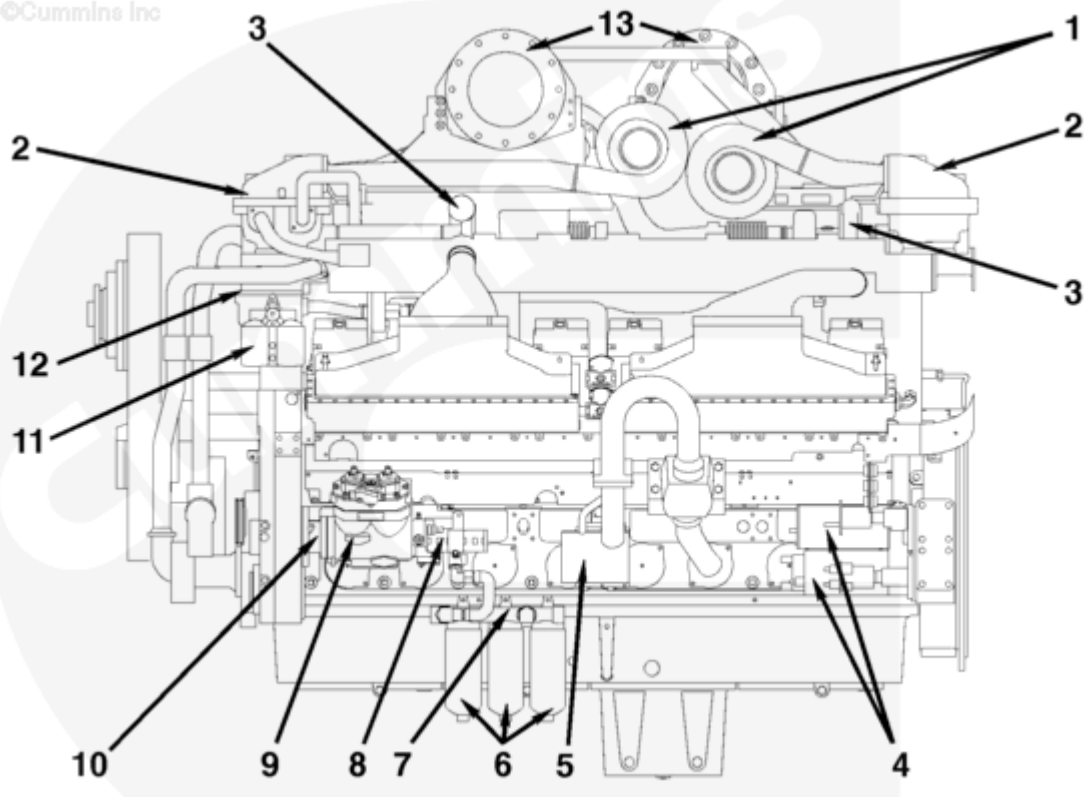
1. Intercooler assembly
2. Air crossover connection
3. Aftercooler assembly
4. Flywheel housing
5. Eliminator full-flow/bypass filtration system
6. Flex plate
7. Cylinder block
8. Rocker lever housing
9. Rocker lever cover
10. Turbocharger
11. Exhaust outlet flange.



19600232

QSK60, 2-Stage - Right View

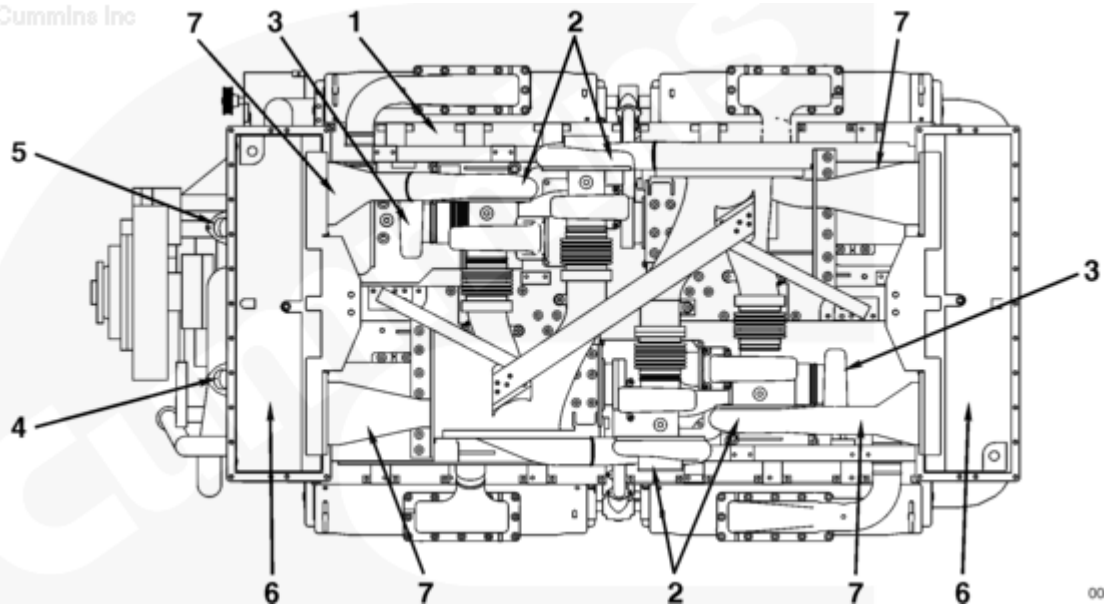
1. Low-pressure turbochargers
2. Intercooler assemblies
3. High-pressure turbochargers
4. Water bypass tube
5. Water pump drive
6. Water pump
7. Water pump inlet
8. Eliminator full-flow/bypass filtration system
9. Oil pan sump
10. Oil pan adapter plate
11. Oil pan adapter
12. Flywheel housing
13. Engine oil level dipstick
14. Oil fill tube
15. Cam follower cover
16. Aftercooler assembly.



19600233

QSK60, 2-Stage - Left View

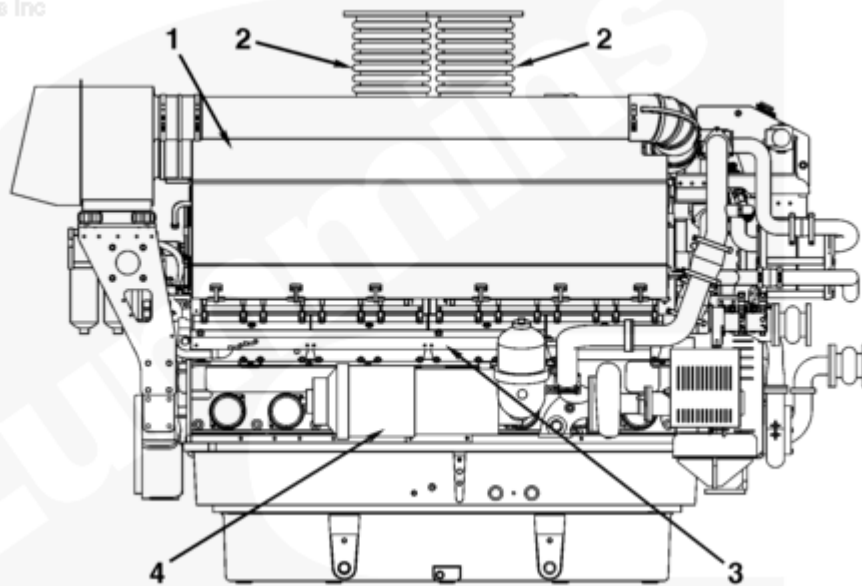
1. Low-pressure turbochargers
2. Intercooler assemblies
3. High-pressure turbochargers
4. Starters
5. Electronic control module (ECM)
6. Fuel filters
7. Fuel filter head
8. Fuel pump
9. Air compressor
10. Compressor drive
11. Coolant filters
12. Thermostat housing
13. Exhaust outlet flange.



00600025

QSK60, 2-Stage - Top View

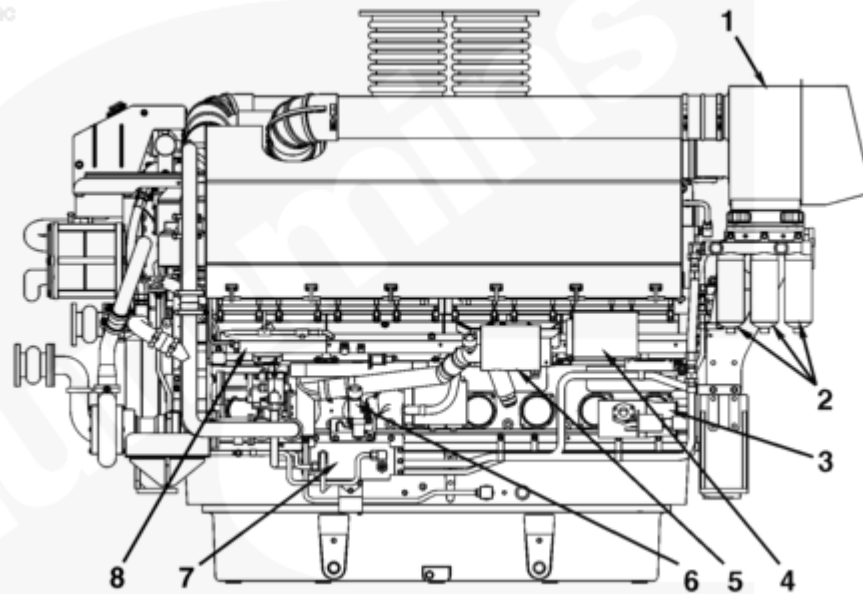
1. Rocker lever cover
2. Low-pressure turbochargers
3. High-pressure turbochargers
4. Left-bank water outlet
5. Right-bank water outlet
6. Intercoolers
7. Air crossovers into intercoolers.



00600215

QSK60 Marine - Right Side View

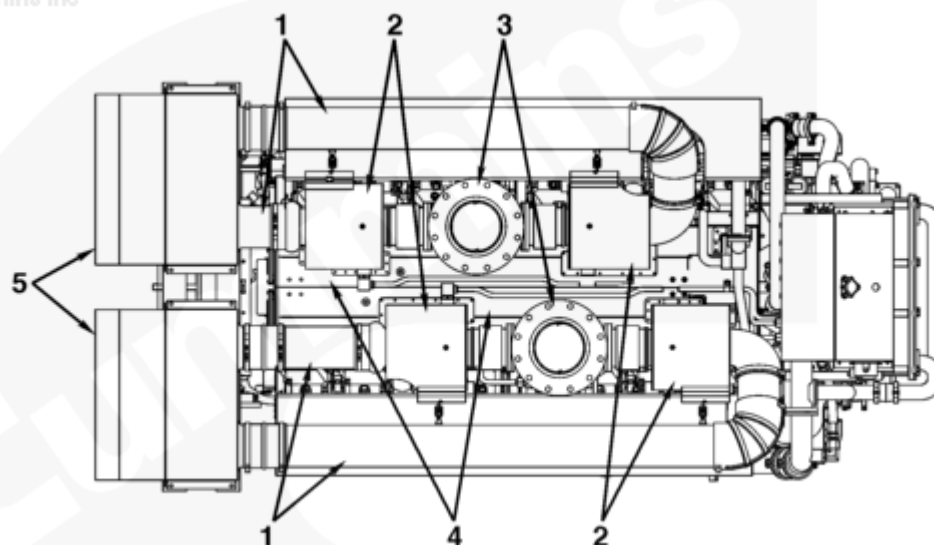
1. Aftercooler heat shield
2. Exhaust outlet bellows
3. Right-bank wiring harness
4. Eliminator lubricating oil filter assembly.



00600216

QSK60 Marine - Left Side View

1. Air cleaner assembly
2. Fuel filters
3. Starter
4. Electronic control module (ECM)
5. Crankcase breather
6. Oil fill and dipstick tubes
7. Fuel cooler (optional)
8. Left-bank wiring harness



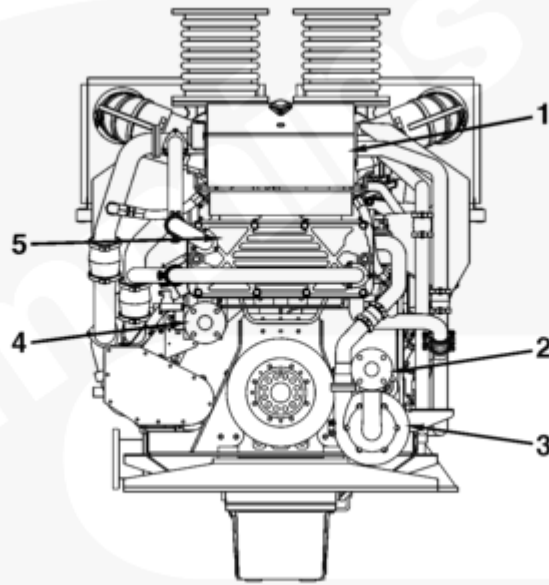
00600217

QSK60 Marine - Top View

1. Air intake tubes
2. Turbocharger heat shields
3. Exhaust outlet connections

4. Exhaust manifold heat shields
5. Air cleaner assemblies.

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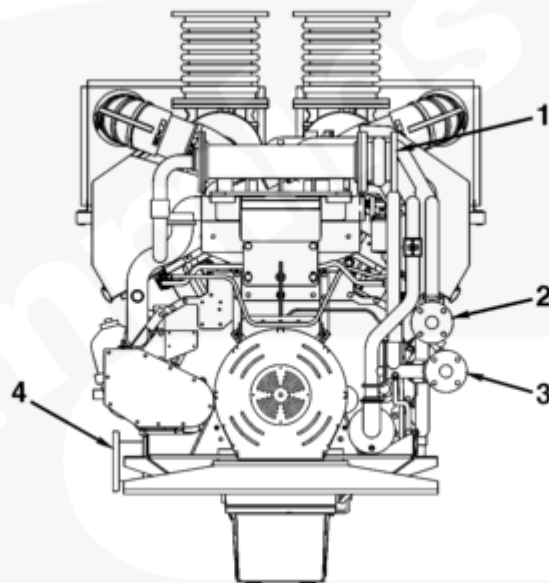


00600213

QSK60 Marine Heat Exchanger Cooled - Front View

1. Expansion tank
2. Seawater inlet connection
3. Seawater pump
4. Seawater outlet connection
5. Heat exchanger.

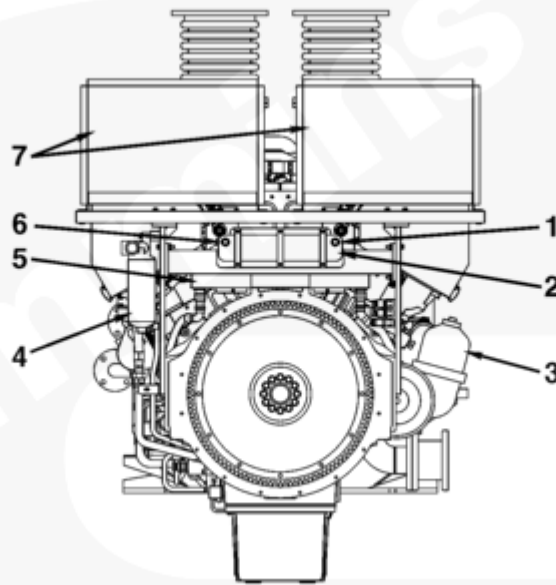
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00600212

QSK60 Marine Keel Cooled - Front View

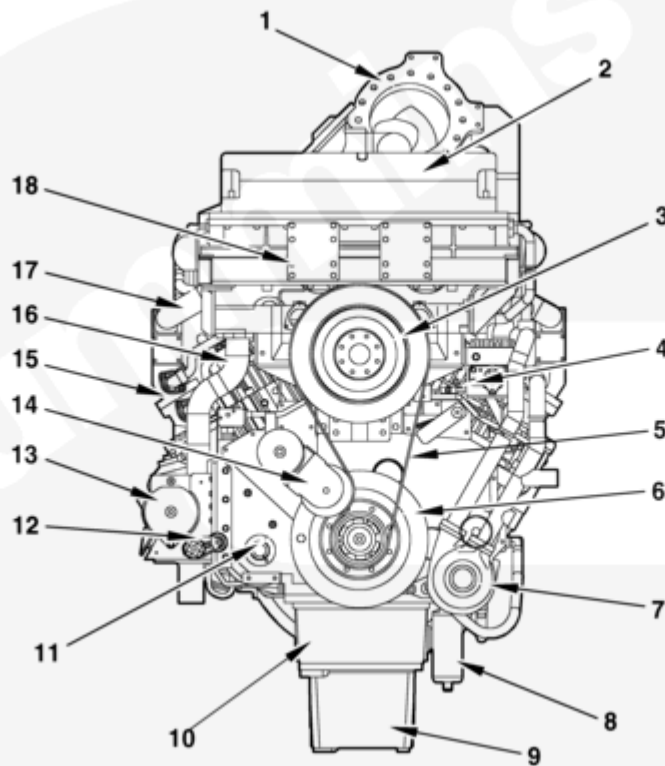
1. Engine coolant outlet connection to main engine cooler
2. LTA coolant outlet connection to LTA cooler
3. LTA coolant inlet connection from LTA cooler
4. Engine coolant inlet connection from main engine cooler.



00600214

QSK60 Marine - Rear View

1. Gear oil inlet or outlet connection
2. Gear oil cooler (optional)
3. Eliminator lubricating oil filter assembly
4. Fuel filter
5. Rear engine harness
6. Gear oil inlet or outlet connection
7. Air filter assemblies.

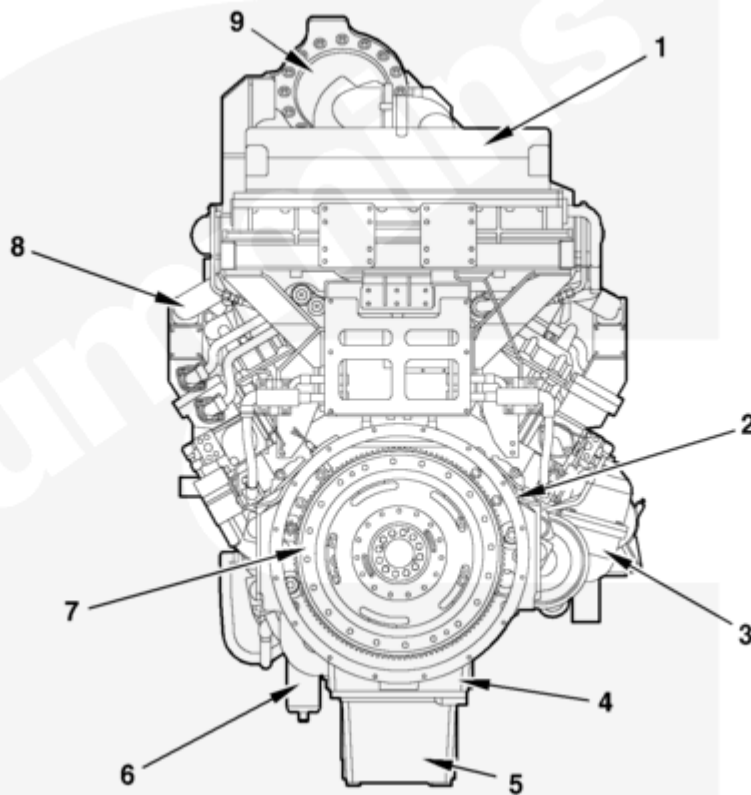


00600060

QSK78 - Front View

1. Exhaust outlet flange
2. Intercooler assembly
3. Fan hub assembly
4. LTA thermostat assembly block
5. Drive belt
6. Vibration damper
7. LTA water pump
8. Fuel filter
9. Oil pan sump
10. Oil pan adapter
11. Front gear cover
12. Alternator drive belt tensioner assembly
13. Alternator
14. Drive belt tensioner assembly
15. Aftercooler assembly
16. Coolant bypass tube
17. Air crossover connection
18. Front alignment plate.

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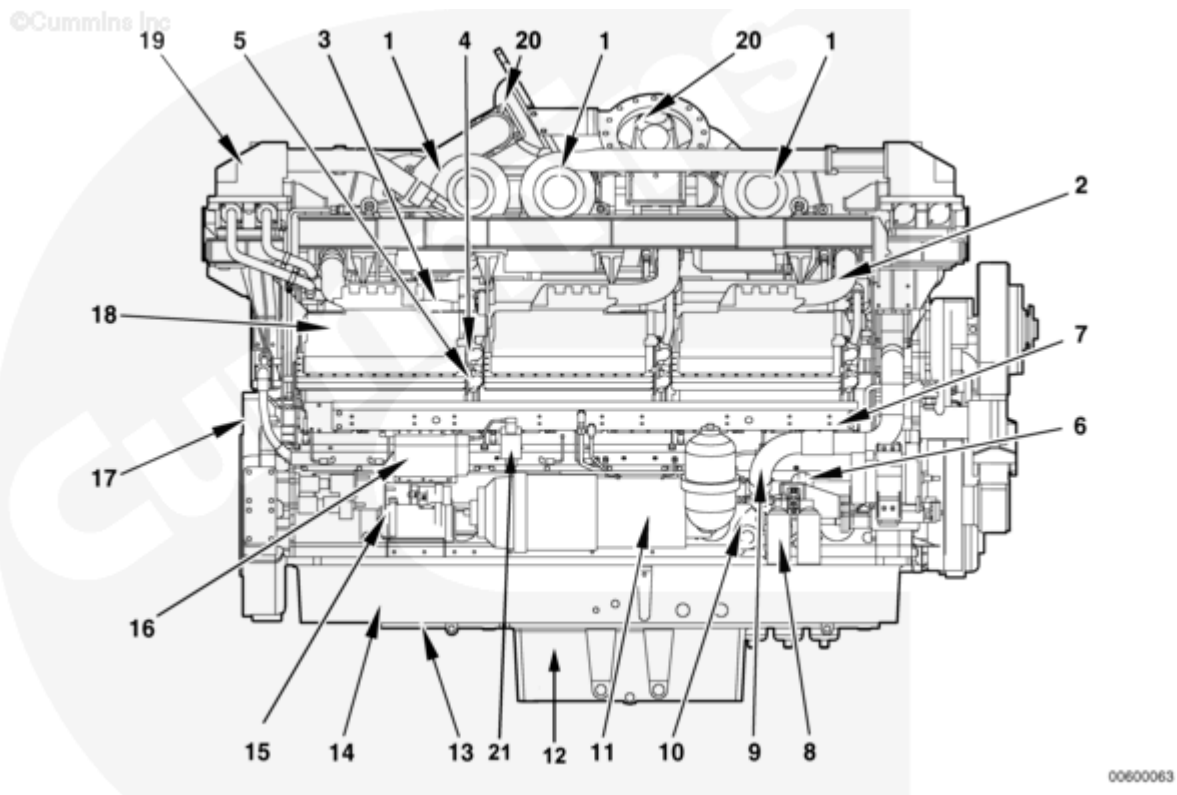


19401001

QSK78 - Rear View

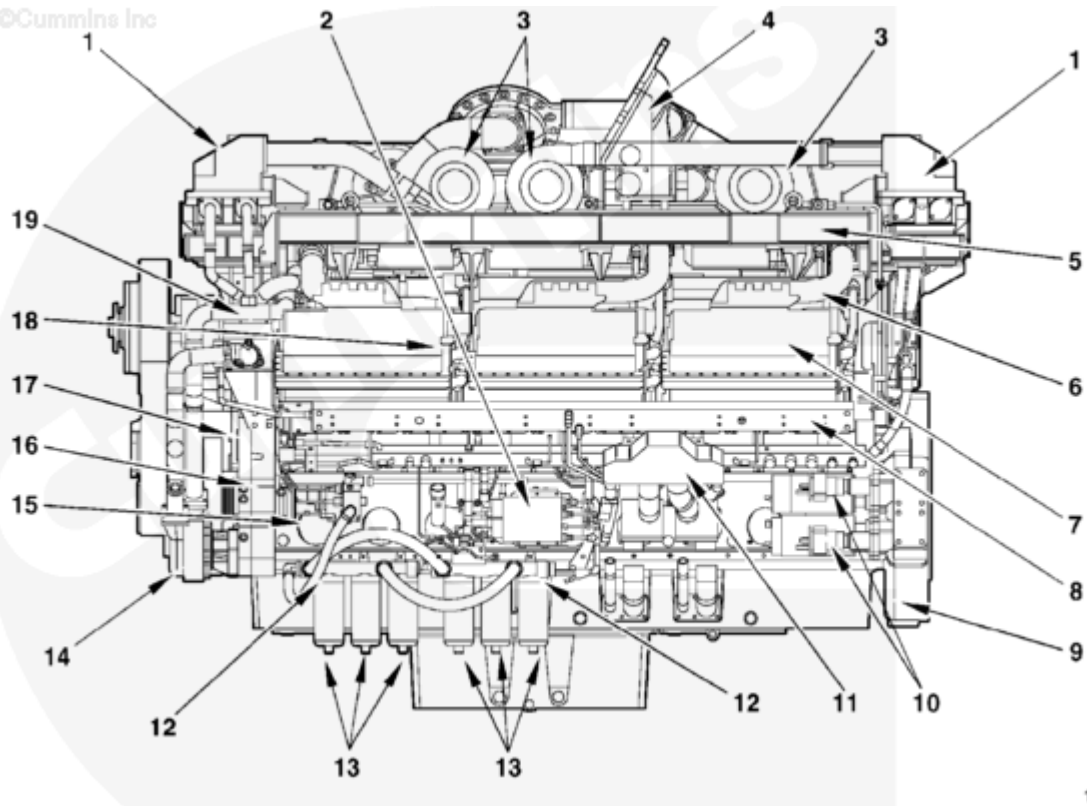
1. Intercooler assembly
2. Flywheel housing
3. Eliminator™ full-flow/bypass filtration system
4. Oil pan adapter
5. Oil pan sump
6. Fuel filter

- 7. Flywheel
- 8. Aftercooler assembly
- 9. Exhaust outlet flange.



QSK78 - Right View

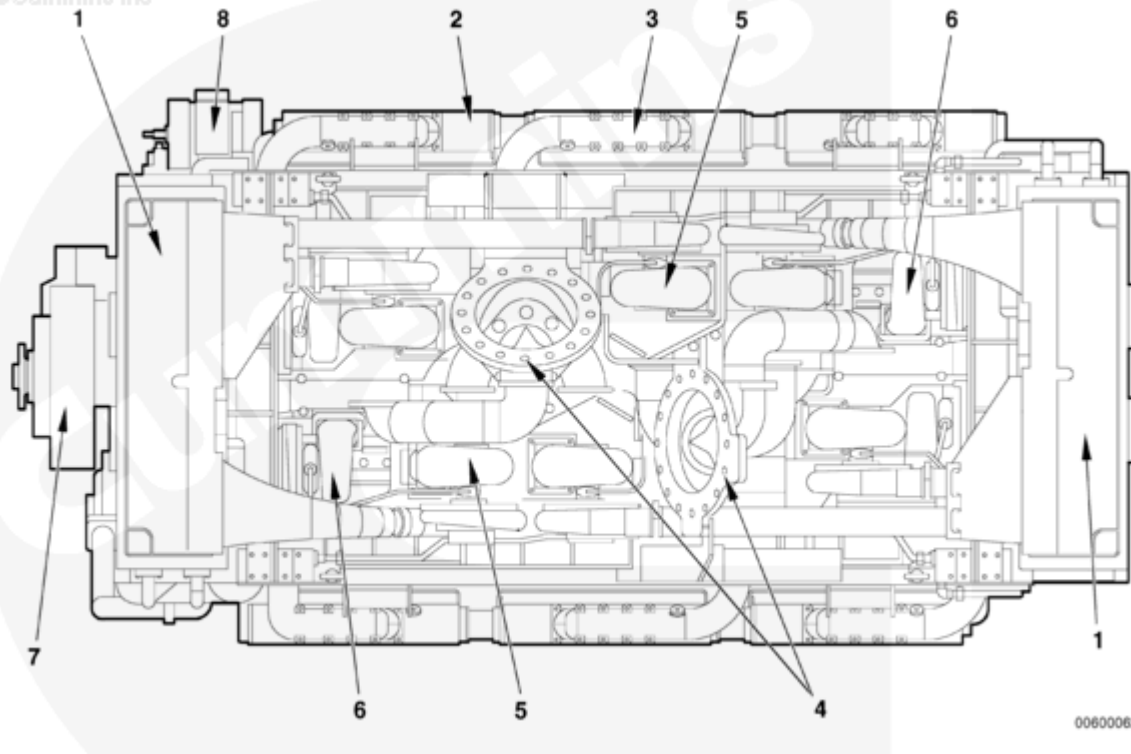
- 1. Low pressure turbochargers
- 2. Air crossover connections
- 3. Rocker lever cover
- 4. Aftercooler water outlet tube
- 5. Aftercooler water inlet tube
- 6. Water pump
- 7. Fuel manifold
- 8. Coolant filters
- 9. Coolant bypass tube
- 10. Water inlet connection
- 11. Eliminator™ oil filter
- 12. Oil pan sump
- 13. Oil pan adapter plate
- 14. Oil pan adapter
- 15. PreLub™ starter system
- 16. CENSE™ module
- 17. Flywheel housing
- 18. Aftercooler assembly
- 19. Intercooler assembly
- 20. Exhaust outlet flange.



19401002

QSK78 - Left View

1. Intercooler assemblies
2. Electronic control module (ECM)
3. Low pressure turbochargers
4. Exhaust outlet flange
5. 2-stage frame
6. Air crossover connection
7. Aftercooler assemblies
8. Fuel manifold
9. Flywheel housing
10. Starter motors
11. Crankcase breather assembly
12. Fuel filter heads
13. Fuel filters
14. LTA water pump
15. Fuel pump
16. Front gear housing
17. Front gear cover
18. Aftercooler water outlet tube
19. LTA thermostat housing block.

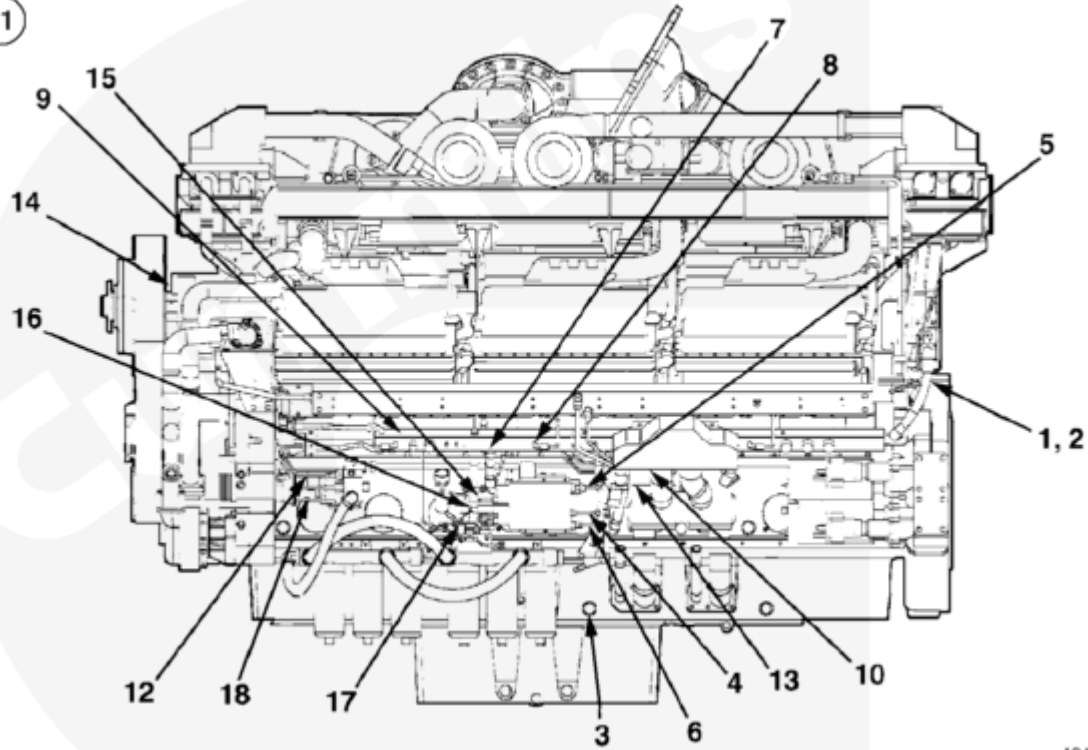


00600062

QSK78 - Top View

1. Intercooler assemblies
2. Aftercooler assemblies
3. Air crossover connections
4. Exhaust outlet flange
5. Low pressure turbochargers
6. High pressure turbochargers
7. Fan hub assembly
8. Alternator.

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QSK78 - Left View (sensor locations)

1. Oil pressure sensor, main rifle
2. Speed sensor
3. Analog oil level sensor
4. Fuel temperature sensor
5. Timing pressure sensor
6. Fuel rail pressure
7. Barometric pressure sensor
8. Boost pressure sensor
9. Intake air temperature sensor
10. Blowby pressure sensor
11. Coolant level sensor (off engine)
12. Fuel pump pressure sensor
13. Coolant pressure sensor
14. Coolant temperature sensor (behind fan hub)
15. Timing actuators (2)
16. Fuel shutoff valve
17. Rail actuator
18. Fuel pump actuator.

